

Usnea VCF filtering

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VCF QC viz for Usnea resequencing data

35 *Usnea florida* and *Usnea subfloridana* resequencing samples were aligned to the *U. subfloridana* genome using BWA-MEM2, and SNPs were called using Freebayes. Here, 53,653 SNPs were randomly subsampled out of 4,496,909 SNPs for QC purposes. VCFtools was used to calculate several QC metrics visualized in ggplot below.

Setup

Load tidyverse and set the working directory.

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages ————— tidyverse 2.0.0 —
## ✓ dplyr      1.2.1      ✓ readr      2.2.0
## ✓ forcats    1.0.1      ✓ stringr    1.5.1
## ✓ ggplot2    4.0.3      ✓ tibble     3.3.1
## ✓ lubridate  1.9.5      ✓ tidyr      1.3.2
## ✓ purrr      1.0.2
## — Conflicts ————— tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
setwd("/home/FM/sblanco/GWAS_8072026/usub/04_freebayes/QC")
```

Variant Quality

Phred quality score for each site. ~24,000 SNPs have QUAL < 30 (~50%).

```
var_qual <- read_delim("usub_vcf_subset.lqual", delim = "\t",
                      col_names = c("chr", "pos", "qual"), skip = 1)
```

```
## Rows: 53653 Columns: 3
## — Column specification —————
## Delimiter: "\t"
## chr (1): chr
## dbl (2): pos, qual
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
summary(var_qual$qual)
```

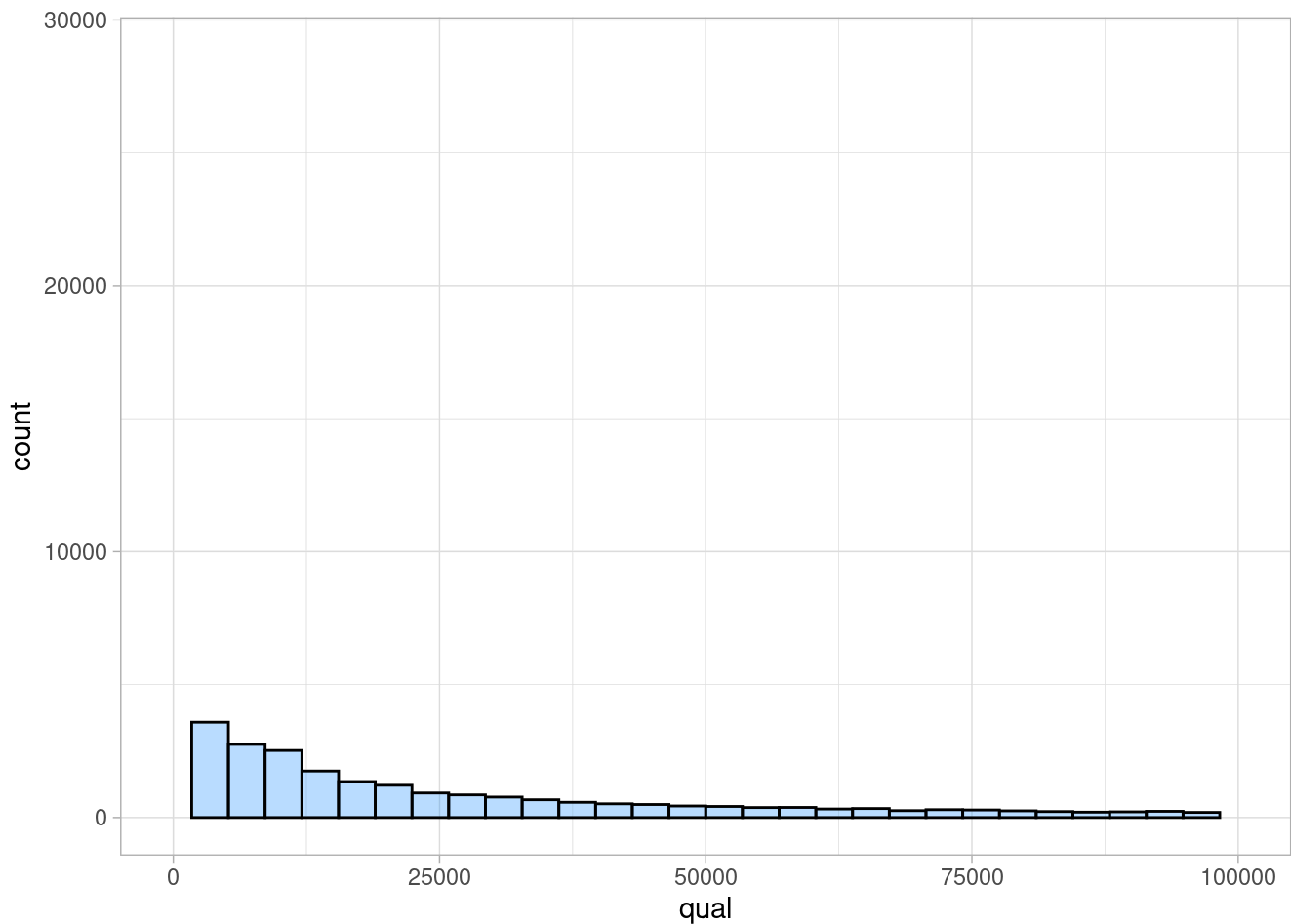
```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
##      0.0      0.0      647.9   17409.7   17878.9 1460400.0
```

```
a <- ggplot(var_qual, aes(qual)) +
  geom_histogram(fill = "dodgerblue1", colour = "black", alpha = 0.3)
a + theme_light() + xlim(0, 100000)
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
## Warning: Removed 2498 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_bar()`).
```



Decision: set min QUAL to 30.

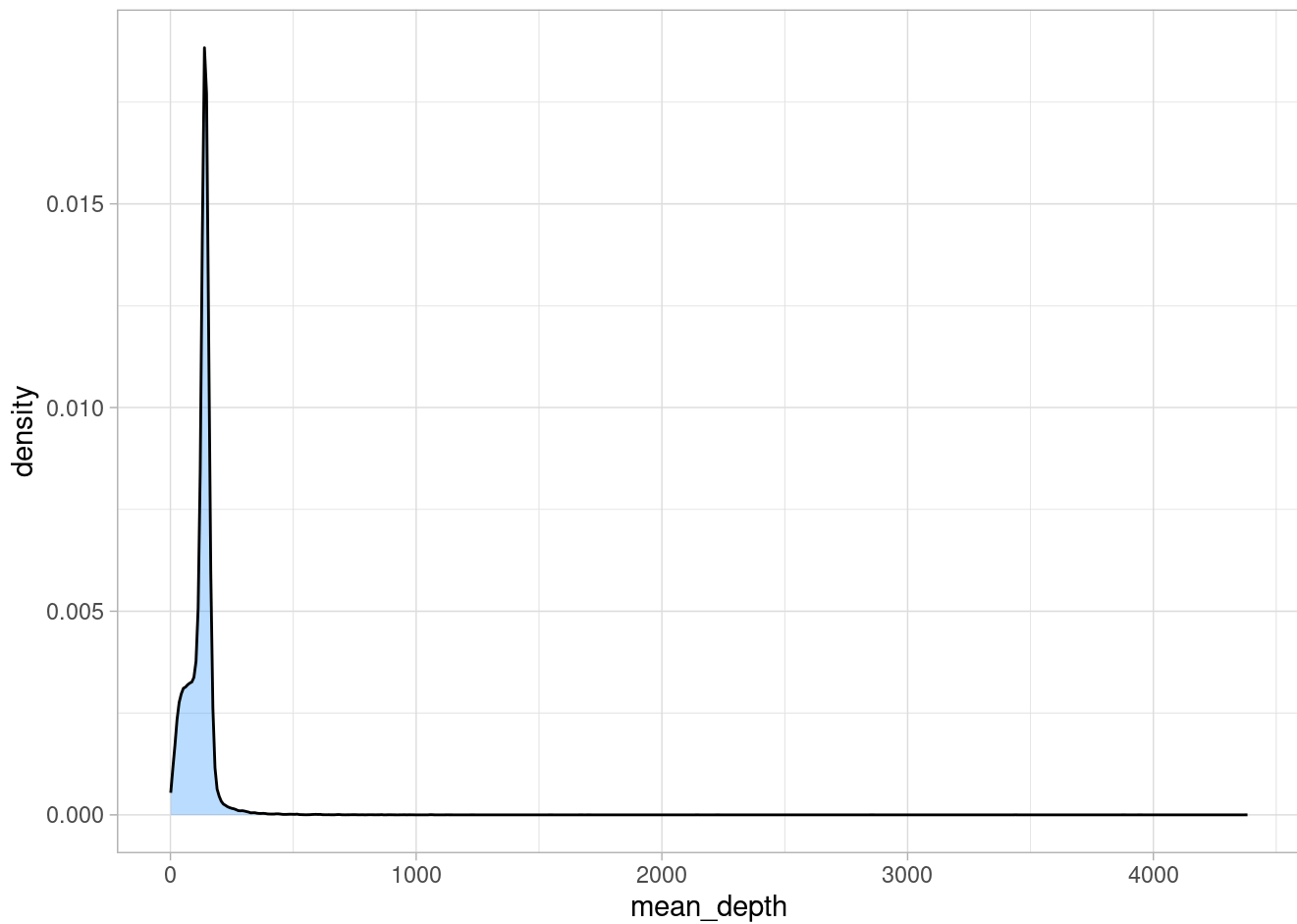
Variant Mean Depth

Mean number of reads mapped to each position.

```
var_depth <- read_delim("usub_vcf_subset.ldepth.mean", delim = "\t",  
                        col_names = c("chr", "pos", "mean_depth", "var_depth"), skip =  
                        1)
```

```
## Rows: 53653 Columns: 4  
## — Column specification —————  
## Delimiter: "\t"  
## chr (1): chr  
## dbl (3): pos, mean_depth, var_depth  
##  
## i Use `spec()` to retrieve the full column specification for this data.  
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
a <- ggplot(var_depth, aes(mean_depth)) +  
  geom_density(fill = "dodgerblue1", colour = "black", alpha = 0.3)  
a + theme_light()
```

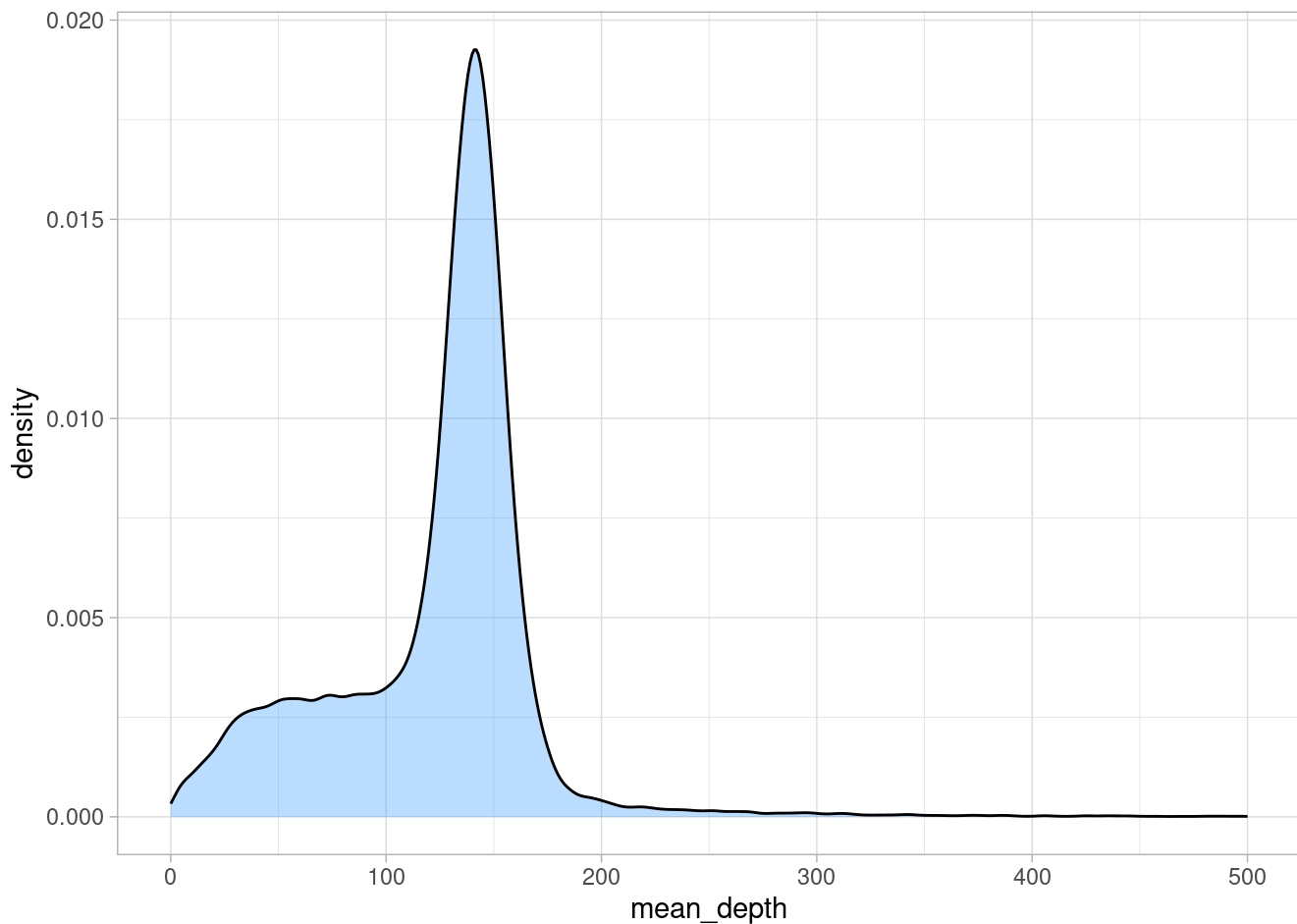


```
summary(var_depth$mean_depth)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.25  100.89  134.69  124.52  147.83 4382.97
```

```
a + theme_light() + xlim(0, 500)
```

```
## Warning: Removed 116 rows containing non-finite outside the scale range
## (`stat_density()`).
```



Decision: set min depth to 10, max depth to $\sim 2 \times \text{mean}$ ($124.52 \times 2 \approx 250$). This handles outliers that could reflect paralogous or repetitive regions.

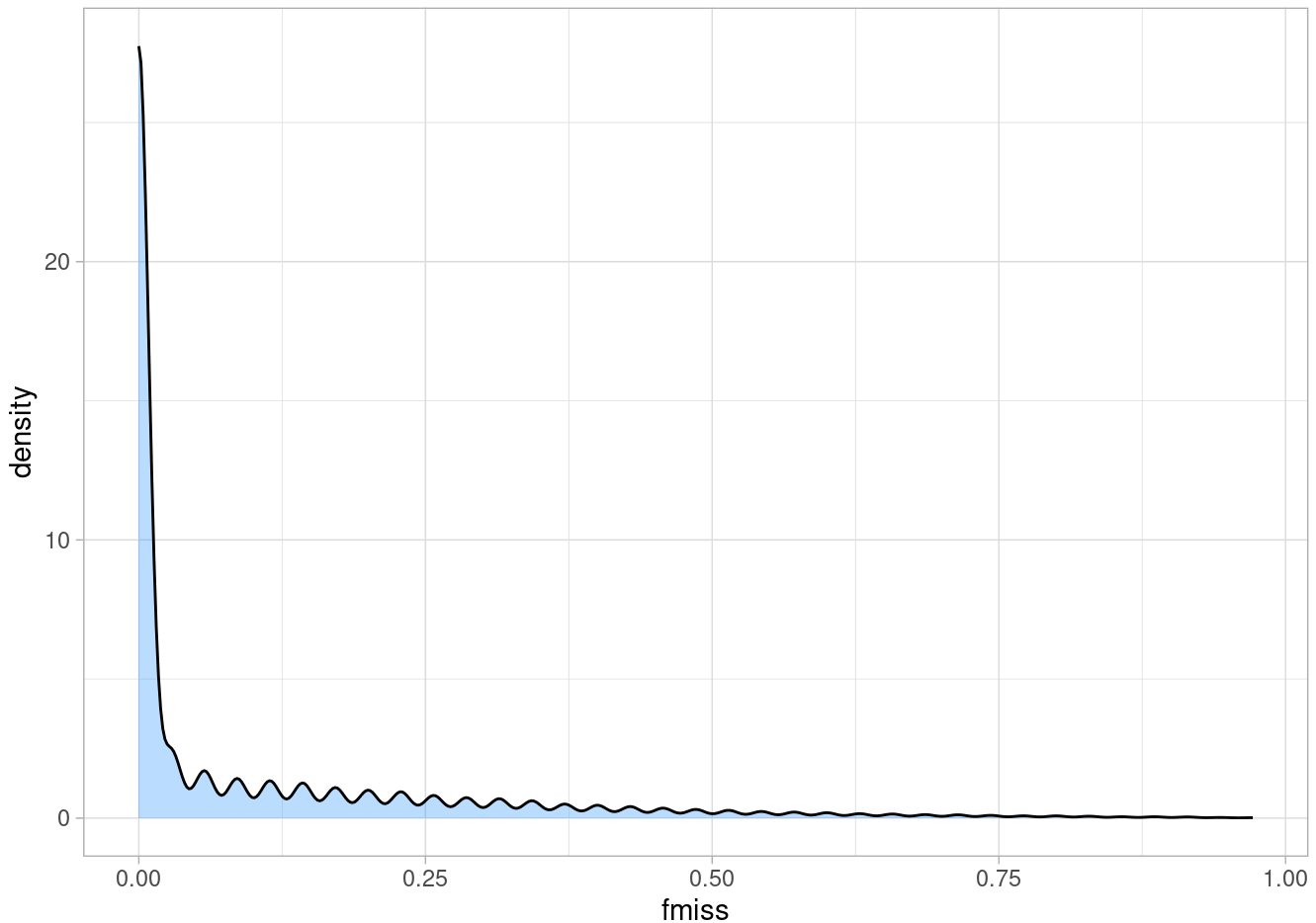
Variant Missingness

Proportion of individuals lacking a genotype at each call site.

```
var_miss <- read_delim("usub_vcf_subset.lmiss", delim = "\t",
                      col_names = c("chr", "pos", "nchr", "nfiltered", "nmiss", "fmiss"), skip = 1)
```

```
## Rows: 53653 Columns: 6
## — Column specification —————
## Delimiter: "\t"
## chr (1): chr
## dbl (5): pos, nchr, nfiltered, nmiss, fmiss
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
a <- ggplot(var_miss, aes(fmiss)) +
  geom_density(fill = "dodgerblue1", colour = "black", alpha = 0.3)
a + theme_light()
```



```
summary(var_miss$fmiss)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.09253 0.11429 0.97143
```

Decision: Typically 75–95% completeness is used as a threshold. Going with 90% (MISS = 0.10).

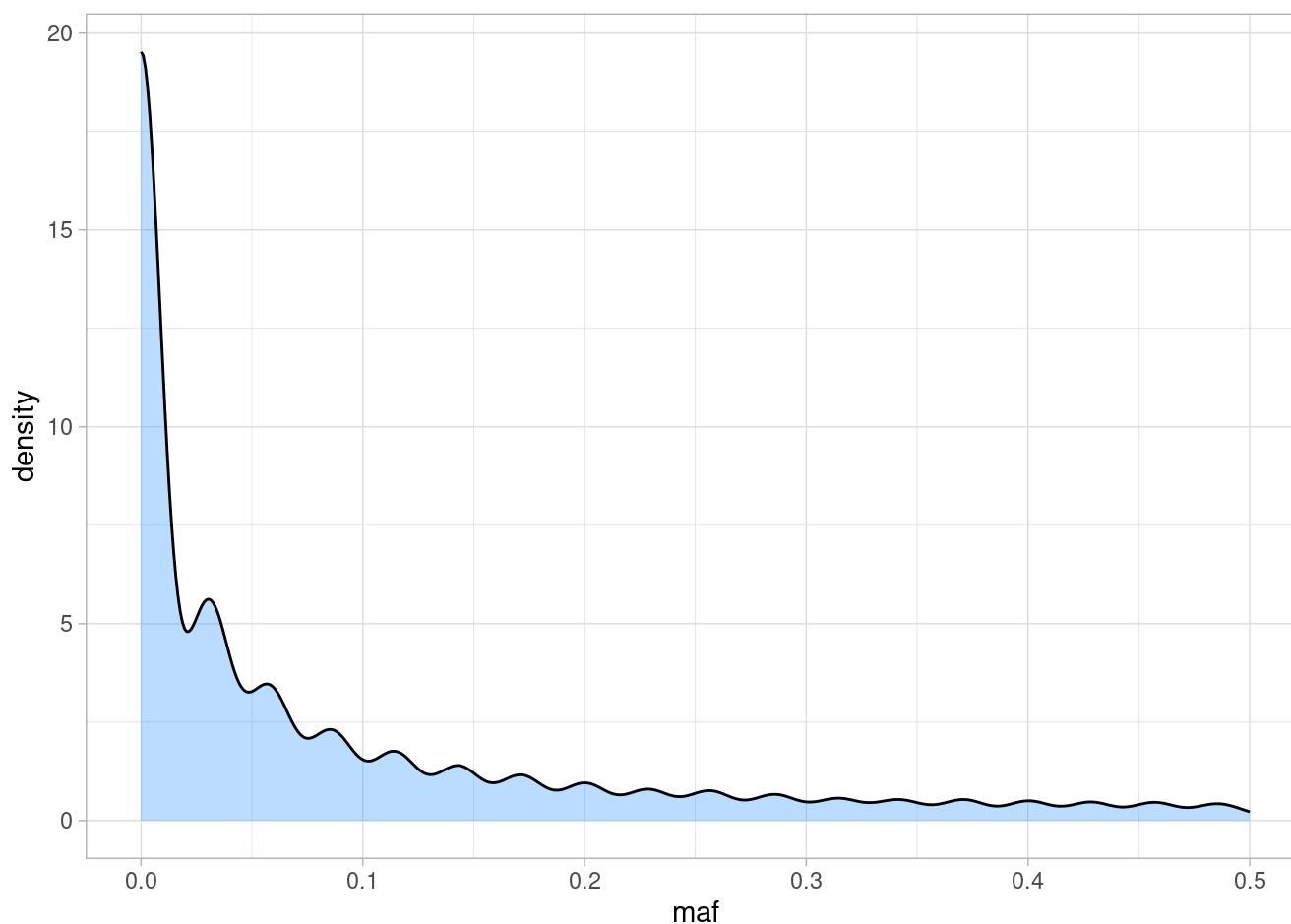
Minor Allele Frequency

```
var_freq <- read_delim("usub_vcf_subset.frq", delim = "\t",
  col_names = c("chr", "pos", "nalleles", "nchr", "a1", "a2"), ski
p = 1)
```

```
## Rows: 41908 Columns: 6
## — Column specification —————
## Delimiter: "\t"
## chr (1): chr
## dbl (5): pos, nalleles, nchr, a1, a2
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
var_freq$maf <- var_freq %>%
  select(a1, a2) %>%
  apply(1, function(z) min(z))

a <- ggplot(var_freq, aes(maf)) +
  geom_density(fill = "dodgerblue1", colour = "black", alpha = 0.3)
a + theme_light()
```



```
summary(var_freq$maf)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.02857 0.08322 0.11429 0.50000
```

Decision: A large number of variants have low-frequency alleles. Will set MAF threshold to 0.1.

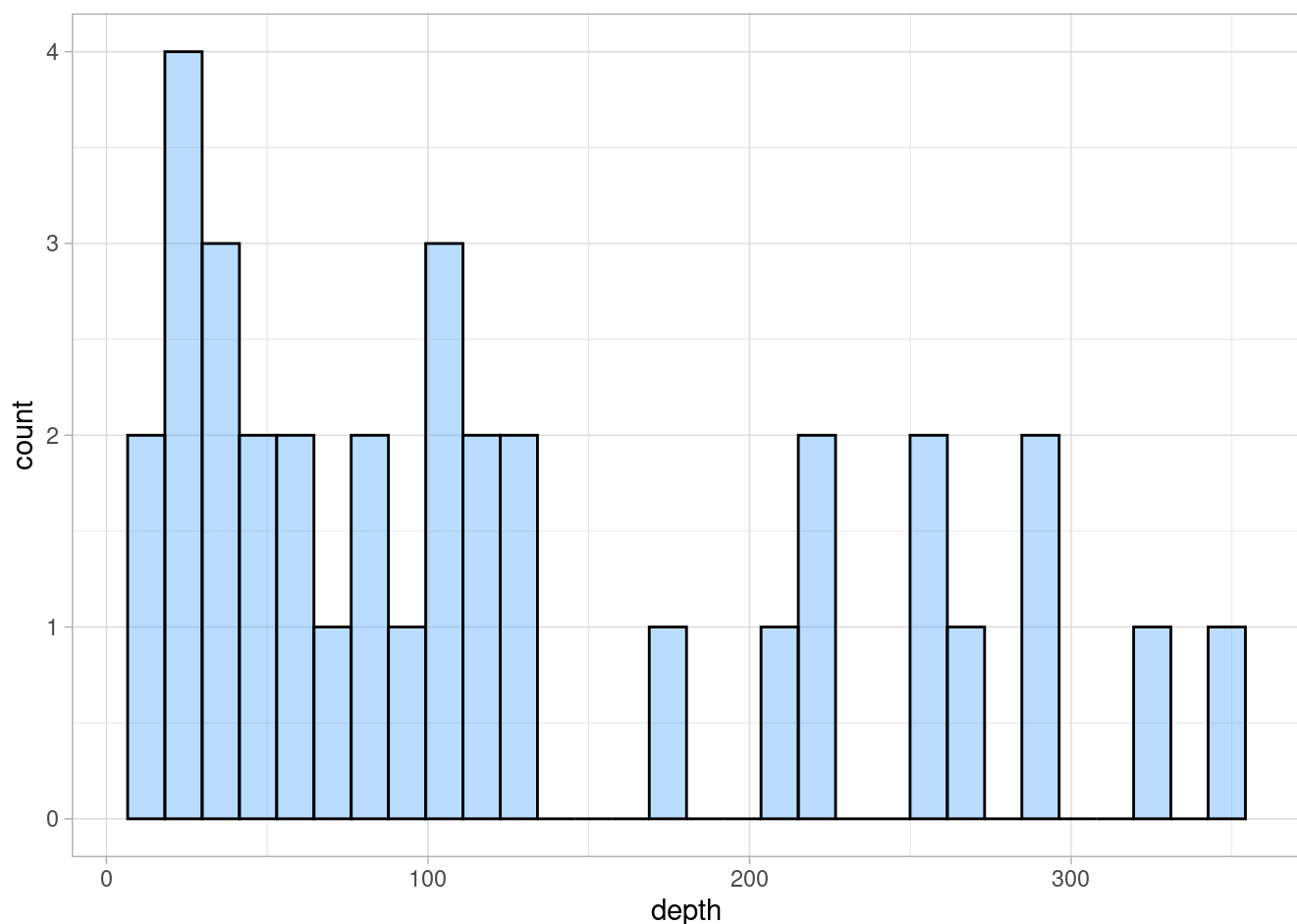
Mean Depth Per Individual

```
ind_depth <- read_delim("usub_vcf_subset.iddepth", delim = "\t",  
                        col_names = c("ind", "nsites", "depth"), skip = 1)
```

```
## Rows: 35 Columns: 3  
## — Column specification —————  
## Delimiter: "\t"  
## chr (1): ind  
## dbl (2): nsites, depth  
##  
## i Use `spec()` to retrieve the full column specification for this data.  
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
a <- ggplot(ind_depth, aes(depth)) +  
  geom_histogram(fill = "dodgerblue1", colour = "black", alpha = 0.3)  
a + theme_light()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
summary(ind_depth$depth)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      12.38   40.14  100.71  126.47  216.73  348.40
```

To note: Sequencing depth ranges from 12 to 348 — a bit of spread there.

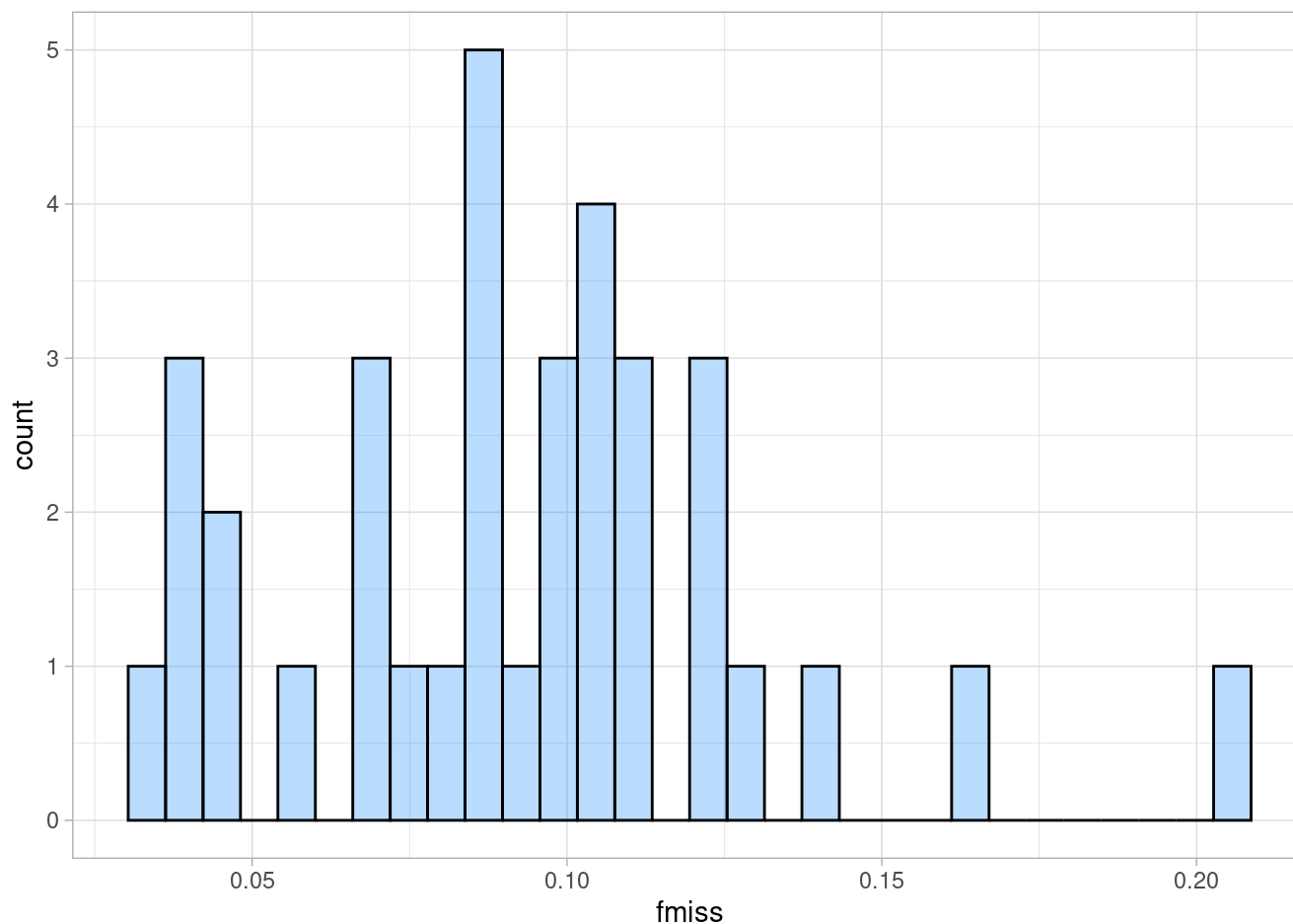
Proportion of Missing Data Per Individual

```
ind_miss <- read_delim("usub_vcf_subset.imiss", delim = "\t",
                      col_names = c("ind", "ndata", "nfiltered", "nmiss", "fmiss"), skip = 1)
```

```
## Rows: 35 Columns: 5
## — Column specification —————
## Delimiter: "\t"
## chr (1): ind
## dbl (4): ndata, nfiltered, nmiss, fmiss
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
a <- ggplot(ind_miss, aes(fmiss)) +  
  geom_histogram(fill = "dodgerblue1", colour = "black", alpha = 0.3)  
a + theme_light()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
summary(ind_miss$fmiss)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   
## 0.03329 0.07042 0.09271 0.09253 0.11020 0.20567
```

To note: Missingness per individual ranges from 0.03 to 0.20.